

Parusan Natheeswaran

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SUMMARY OF QUALIFICATIONS

- Computer Engineering co-op student with 2 years of ML engineering at Outlier AI in speech and conversational AI.
- Ships across the stack: offline AI desktop apps, tool-calling AI agents, RISC-V CPU cores, and GPU LLM inference.
- Builds tested, documented work: 900+ automated tests, formal verification, CI/CD, and written design specs.
- Collaborative communicator who shipped a feature to millions at Riot Games and leads 10+ staff as a shift manager.

EDUCATION

Toronto Metropolitan University

Sep. 2023 – Present

Bachelor of Engineering, Computer Engineering (Co-op)

Toronto, ON

- **Co-op Availability:** January 2027 (12 months) or May 2027 (8, 12, or 16 months)
- **Relevant Coursework:** Software Systems, Digital Systems, Operating Systems, Algorithms & Data Structures

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, Go, SQL, SystemVerilog, Bash

AI/ML: PyTorch, AMD ROCm, LLMs, AI agents, tool calling, RAG, QLoRA fine-tuning, Claude API, Ollama, whisper.cpp

Software & Cloud: React, Electron, Node.js, Flask, FastAPI, REST APIs, PostgreSQL, Redis, Docker, AWS, Azure, Fly.io

Hardware: RISC-V, CPU pipelines, FPGA, formal verification (SymbiYosys), cocotb, Verilator, Yosys/nextpnr

Tools & Process: Git, GitHub Actions CI/CD, pytest, Vitest, Playwright, Linux, JIRA, Agile Scrum, code reviews, Figma

WORK EXPERIENCE

Machine Learning Engineer

Oct. 2024 – Present

Outlier AI

Oakland, CA (Remote)

- Improved accuracy and reliability across 20+ AI/ML projects by developing and optimizing speech recognition and conversational AI models in Python, C++, and Java.
- Earned 70+ acknowledgments by owning end-to-end workflows, from exploratory data analysis to model deployment.

Game Developer

May 2024 – Aug. 2024

Riot Games

Los Angeles, CA (Remote)

- Shipped a playable character to millions of users on a 3-month deadline, prototyping features in Python.
- Cut crash rates 15% by triaging and resolving 50+ critical bugs with design, engineering, and QA teams.

Shift Manager

Dec. 2025 – Present

Town of Milton

Milton, ON

- Lead 10+ staff per shift, handling escalations, safety oversight, process compliance, and weekly status reports.

PROJECTS

Xenolinguist | *TypeScript, React 19, Electron, Node.js, Ollama, whisper.cpp* | [Download](#)

Apr. 2026 – Present

- Shipped v1.0.0 of an offline AI research partner for decoding unknown languages: a Windows desktop app with auto-update that runs whisper.cpp, wav2vec2 IPA recognition, and local LLMs entirely on-device.
- Routed each task to a heavy or light Ollama model with streamed responses, grounded the AI chat in the user's own dictionary and grammar, and labelled phonetic guesses apart from real transcriptions.
- Designed a six-phase workflow from raw audio to live translation, with number-base auto-detection, 0–100 confidence scores, a Ctrl+K command palette, and an AI-generated practice-language sandbox.

Bingery | *React, TypeScript, Python, Flask, Claude API, Fly.io* | [Live Demo](#)

Mar. 2026 – Present

- Built a live anime discovery PWA with a Claude tool-calling AI agent (6 custom tools, cloud or local models), an IDF-weighted similarity engine, and a 75-route JWT-secured REST API.
- Backed it with 690+ TDD-first tests in GitHub Actions CI/CD, daily jobs syncing a 14,000+ title catalog, and a mobile rework driven by usability testing with 15+ users.

Pulsar-V | *SystemVerilog, RISC-V, FPGA, cocotb, SymbiYosys, Yosys/nextpnr*

Jul. 2026 – Present

- Designed two RV32I CPU cores, including a 5-stage pipeline with forwarding and a BTB branch predictor (83.8–99.6% accuracy, 1.00–1.27 CPI), closing timing at 38.8 MHz on a Lattice ECP5 FPGA.
- Verified both cores with 43/43 riscv-formal proofs, 38/38 riscv-arch-test against the SAIL model, and 41/41 riscv-tests.

EdgeSpark | *Python, PyTorch, AMD ROCm, Radeon RX 7900 XTX*

Apr. 2026 – Present

- Speculative decoding on one AMD GPU: exact Qwen3-4B output, 9.5 GB VRAM, +43% modelled throughput.

Parley | *Python, whisper.cpp, Vulkan, Ollama*

Jul. 2026

- Extended an open-source dictation tool into an offline voice-to-text app with a 135 ms GPU round-trip and 186 tests.